

Chemistry Detailed Formula Book

Important formulas, relations and quick-use guide

Mole & Concentration

- $n = m / M$ (moles = given mass / molar mass)
- $N = n \times N_A$ (number of particles)
- $M = n / V$ (molarity, with V in litres)
- $M_1V_1 = M_2V_2$ (dilution)
- moles of gas = PV / RT

Atomic Structure

- $A = Z + N$
- Number of neutrons $N = A - Z$
- Maximum electrons in a shell = $2n^2$
- Subshell capacities: s = 2, p = 6, d = 10, f = 14

Gaseous State

- $PV = nRT$
- $P_1V_1 = P_2V_2$ at constant T
- $V_1/T_1 = V_2/T_2$ at constant P
- $P_1V_1/T_1 = P_2V_2/T_2$ for a fixed amount of ideal gas

Thermodynamics

- $\Delta H = H(\text{products}) - H(\text{reactants})$
- $\Delta G^\circ = -RT \ln K$
- At constant pressure, heat exchanged is related to enthalpy change
- Hess's law: overall ΔH is the sum of the enthalpy changes of component steps

Solutions

- $M = n/V$
- Mass percentage = (mass of solute / mass of solution) $\times 100$
- Molality = moles of solute / kg of solvent
- $M_1V_1 = M_2V_2$ for dilution

Electrochemistry

- Oxidation occurs at anode
- Reduction occurs at cathode
- Cell potential is obtained from cathode and anode potentials according to the chosen convention
- Faraday-type numerical problems require careful use of charge, electrons and molar quantities

Chemical Kinetics

- Rate describes concentration change with time
- For a first-order process, $\ln(N_0/N)$ is proportional to time
- Half-life for a first-order process is independent of initial concentration

Quick Formula Use Guide

- First identify the chapter and the physical quantity asked.
- Convert units before substituting.
- Check whether temperature must be in kelvin.
- Keep significant figures sensible and verify the direction/sign of the answer.